

24-Hour Urinary Sodium and Potassium Excretion and Cardiovascular Risk

Ma Y, He FJ, Sun Q, Yuan C, Kieneker LM, Curhan GC, MacGregor GA, Bakker SJL, Campbell NRC, Wang M, Rimm EB, Manson JE, Willett WC, Hofman A, Gansevoort RT, Cook NR, Hu FB. New England Journal of Medicine. 2022 Jan 20; 386(3):252-263. DOI: 10.1056/NEJMoa2109794 PMID: 34767706

Abstract

BackgroundThe relation between sodium intake and cardiovascular disease remains controversial, owing in part to inaccurate assessment of sodium intake. Assessing 24-hour urinary excretion over a period of multiple days is considered to be an accurate method.**Methods**We included individual-participant data from six prospective cohorts of generally healthy adults; sodium and potassium excretion was assessed with the use of at least two 24-hour urine samples per participant. The primary outcome was a cardiovascular event (coronary revascularization or fatal or nonfatal myocardial infarction or stroke). We analyzed each cohort using consistent methods and combined the results using a random-effects meta-analysis.**Results**Among 10,709 participants, who had a mean ($\pm$ SD) age of 51.5 $\pm$ 12.6 years and of whom 54.2% were women, 571 cardiovascular events were ascertained during a median study follow-up of 8.8 years (incidence rate, 5.9 per 1000 person-years). The median 24-hour urinary sodium excretion was 3270 mg (10th to 90th percentile, 2099 to 4899). Higher sodium excretion, lower potassium excretion, and a higher sodium-to-potassium ratio were all associated with a higher cardiovascular risk in analyses that were controlled for confounding factors ($P \leq 0.005$ for all comparisons). In analyses that compared quartile 4 of the urinary biomarker (highest) with quartile 1 (lowest), the hazard ratios were 1.60 (95% confidence interval [CI], 1.19 to 2.14) for sodium excretion, 0.69 (95% CI, 0.51 to 0.91) for potassium excretion, and 1.62 (95% CI, 1.25 to 2.10) for the sodium-to-potassium ratio. Each daily increment of 1000 mg in sodium excretion was associated with an 18% increase in cardiovascular risk (hazard ratio, 1.18; 95% CI, 1.08 to 1.29), and each daily increment of 1000 mg in potassium excretion was associated with an 18% decrease in risk (hazard ratio, 0.82; 95% CI, 0.72 to 0.94).**Conclusions**Higher sodium and lower potassium intakes, as measured in multiple 24-hour urine samples, were associated in a dose-response manner with a higher cardiovascular risk. These findings may support reducing sodium intake and increasing potassium intake from current levels. (Funded by the American Heart Association and the National Institutes of Health.).

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